Total No. of Questions: 05 Total No. of Printed Pages: 02

Roll No. ________________

BTCS501N

B.Tech/ B.Tech. + M.B.A. / B.Tech. + M.Tech. (CSE, CSE-CF, CSE-BDA, CSE-AIML, CSE-ES, CSE-MA, CSE-BDAI, CSE-CMCI, CSE-AII, CSE-DSI, CSE-FSDI, CSE-ICS, BDCE- Impetus, IT)

V Semester Examination, December 2023

Theory of Computation

Choice Based Credit System (CBCS)

Time: 3 Hrs. **Maximum Marks: 60**

Minimum Pass Marks: 24

Note: (1) All questions carry equal marks, out of which part'A' and 'B' carry 3 marks and part 'C' carries 6 marks.

(2) From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.

(3) Draw neat diagram, wherever necessary.

(4) Assume suitable data wherever necessary.

Q.1(A) Explain the need and uses of NFA in computing. **03**

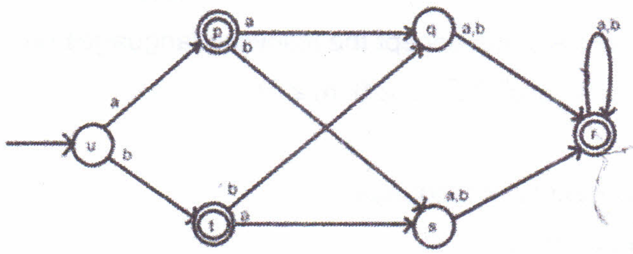
(B) Find a DFA for the following languages on $\Sigma = \{a, b\}$. **03**

$L = \{w : n_a(w) \bmod 3 > n_b(w) \bmod 3\}$

(C) Convert Epsilon NFA to NFA of given language $L = \{ a^n b^m c^k$ where $n,m,k \geq 0\}$. **06**

OR

Minimize of DFA using Myhill-Nerode Theorem.



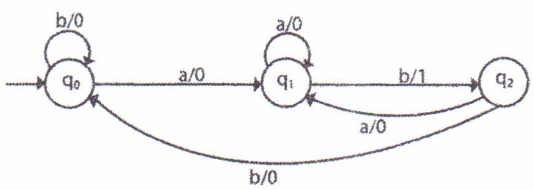
Q.2(A) Explain decision properties of regular languages. **03**

(B) Explain Arden's theorem with taking suitable example. **03**

(C) Let $C_n = \{x \mid x$ is a binary number that is a multiple of $n\}$. Show that for each $n \geq 1$, the language C_n is regular. **06**

OR

Convert following mealy machine to moore machine.



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Q.3(A)	Explain pumping lemma for CFL with taking an example.	**03**
(B)	Draw left derivation tree for L={w € {0, 1}*: w has no pair of consecutive zero }.	**03**
(C)	Let $\Sigma = \{0,1,\#\}$. Let $C = \{x\#x^R\#x \mid x \varepsilon \{0,1\}^*\}$. Show that C' is a CFL.	**06**
	OR	
	Find context-free grammars for the following languages (with $n \geq 0, m \geq 0, k \geq 0$) $L=\{a^n b^m c^k : n + 2m = k\}$.	
Q.4(A)	Explain properties of NPDA.	**03**
(B)	Construct a CFG of a given PDA. M = ({q0,q1}, {0,1}, {X, Zo}, D, q0, Zo, {}) with D (delta): d(q0,0,Z0) = (q0,XZo) d(q0,0,X) = (q0,XX) d(q0,1,X) = (q1,e) d(q1,1,X) = (q1,e) d(q1,e,X) = (q1,e) d(q1,e,Zo) = (q1,e)	**03**
(C)	Give a PDA that recognizes language $B= \{uv \mid u,v \varepsilon \Sigma^*, v\ \varepsilon \Sigma^* 1 \Sigma^* \text{ and } \lvert u \rvert \geq \lvert v \rvert\}$.	**06**
	OR	
	Construct NPDA's that accept the following languages on $\Sigma = \{a, b, c\}$ $L = \{a^n b^{n+m} c^m : n \geq 0, m \geq 1\}$.	
Q.5(A)	Explain decidability of languages.	**03**
(B)	Explain church thesis.	**03**
(C)	Design a turing machine that accepts the Language $L= \{a^n : n \text{ is a prime number}\}$.	**06**
	OR	
	Design a turing machine with no more than three states that accepts the language L(a (a + b)*). Assume that $\Sigma = \{a,b\}$. Is it possible to do this with a two-state machine?	

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